

Realization of Multiple Region-of-Interests in the CCSDS Recommendation for Image Data Compression

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Abstract

For satellite image compression, the Consultative Committee for Space Data Systems (CCSDS) has proposed a processing standard which is so far the most widely implemented in hardware. However, the compression rate is the only factor considered in this standard. For general image compression, there are usually more scalabilities attached, such as region-of-interests (ROI), resolution, component and so on. Above all, ROI is the most popular feature, and JPEG2000 also provides two mechanisms for the ROI coding in the frame-based mode. In our research, a multiple-ROI coding based on CCSDS compression standard is proposed. Instead of using the ROI coding of JPEG2000, the proposed algorithm performs well not only in the frame-based mode but also in the strip-based mode. The latter mode is designed for saving the buffer and decreasing the hardware cost. Besides, the proposed algorithm does not alter the existing image compression standard proposed by CCSDS. That is, it could be implemented in hardware in conjunction with the existing satellite image compression hardware systems without significant modification. In the end, simulation results show that the performance of our method is almost as good as that of JPEG2000 in the frame-based mode, and it still performs well in the strip-based mode.

Keywords: CCSDS, image compression, region-of-interests, scalability, satellite image

1. Introduction

Satellite images have been widely used in many fields such as military affairs, atmospheric sciences, topography recognition, environmental studies, ocean-temperature detection, and so on. An obvious characteristic of satellite images is their large data size. However, users are usually more interested in some regions rather than the full frame of a satellite image. To satisfy this request, region-of-interests coding adopted in image compression is the most popular method. In fact, besides the researches focused on high compression qualities [1], [2], [3], [4], there had been more and more issues concerning about the ROI capability [5], [6], [7]. JPEG2000 standard is one of the popular image compression methods that provides the ROI capability and is very powerful. In the part I of JPEG2000 standard [8], the ROI encoding is implemented with a scaling coefficient method called MAXSHIFT followed by the so-called Embedding Block Coding with Optimal Truncation (EBCOT).

The Consultative Committee for Space Data Systems (CCSDS) is an organization that meets periodically to address data system problems that are common to all participants, including image data compression system [9], [10], [11] and others. Because of the limitation of the hardware used aboard spacecraft, the compression technique described in the recommended standard proposed by CCSDS, which will be denoted as the CCSDS standard hereafter, differs from the JPEG2000 standard in several respects:

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- 1). it specially targets use aboard spacecraft;
- 2). a careful trade-off has been performed between compression performance and complexity;
- 3). being less complex, it can be fully implemented in either hardware or software;
- 4). it has limited set of options, supporting its successful application without in-depth algorithm knowledge.

Because of these characteristics, the CCSDS standard is the most popularly adopted to be implemented in hardware used in satellites or spacecraft. However, the compression rate is the only choice provided in the original CCSDS standard, and this makes this standard weak in respect of functionality.

In this research, we introduce a novel multiple-ROI coding algorithm based on the CCSDS standard, which extends the ROI scalability of the CCSDS standard either in the frame-based mode or in

the strip-based mode. First, we lower down the resolution of a satellite image according to the size of the buffer. Next, the necessary bits for losslessly compressing the ROIs could be estimated by calculating the entropy of the corresponding low-resolution image, and then we allocate the target bits to the ROIs and non ROI properly. In the end, the bit-plane encoding (BPE) is performed to output the bit stream. The proposed method is designed for easy implementation and is compatible with the existing CCSDS recommended standard compression system without significant modification.

The remainder of this paper is organized as follows. A review of popular image compression algorithms that support the ROI capability is given in Section II. In Section III, we shall address our proposed multiple-ROI image compression scheme. Then the simulation results and a further discussion are provided in Section IV. Finally, a brief conclusion is drawn in Section V.

2. Related works

For image compression, the researches concerning about the ROI scalability are more and more popular. Set partitioning in hierarchical trees (SPIHT) algorithm is widely used for image compression since its good compression performance [12], [6], [13],

[14], [15], [16]. In the previous work, an ROI coding algorithm based on SPIHT was proposed by Keun-hyeong Park and Hyun Wook Park. A parent ROI (PROI) mask and a parameter, relative importance (R), were used to perform the ROI coding [6].

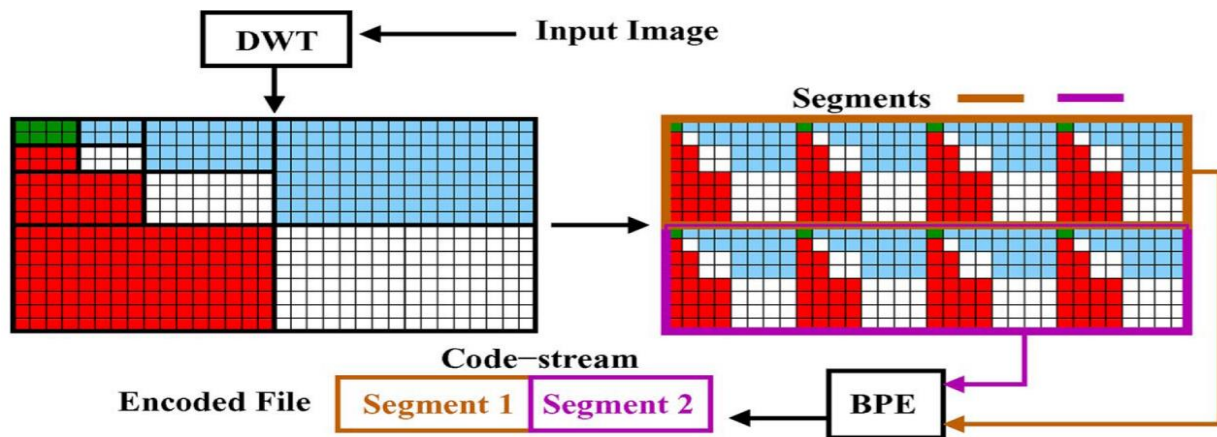


Figure 1: Basic scheme of the CCSDS encoder

JPEG2000 [17], [5], [18], [8] is a powerful coding standard for images in respects of transmission and compression. It provides two methods to realize the ROI coding, one is MAXSHIFT and the other is the general scaling-based method. The bit-planes of the ROI coefficients are shifted upwards properly to ensure that the quality of ROIs in the reconstructed image is the best. Note that MAXSHIFT is a special case of the general scaling-based method.

Another research about ROI scalability based on CCSDS was proposed by F. Garcia-Vlchez [19], where the CCSDS image compression standard was significantly modified. 5-level discrete wavelet transform was used, and the concept of Minimal Unit of informaTion (MUT) was proposed to instead of the gaggle structure in CCSDS standard. Xavier Delaunay [20] also realized the ROI scalability based on the CCSDS compression standard where the MAXSHIFT and scaling-based methods were used. In the scaling-based method, the information of the ROI mask

must be encoded in the compressed bit stream since it is needed in the decoding process.

The encoding process of the CCSDS standard could be roughly divided in three functional parts, as shown in Fig. 1 [19]. The input image is transformed to a coefficient array by a 3-level DWT and then the process, *ReGrouping*, is performed where the transformed coefficients are rearranged in 8×8 blocks. Each block with rearranged coefficients is shown in Fig. 2 [19]. A group of S consecutive blocks are defined as a *segment*. S can be assigned to any positive integer in the range $16 \leq S \leq 2^{20}$, depending on the available buffer used to store the segment. Finally, each segment is independently encoded by bit-plane encoding (BPE). Every 16 consecutive blocks within a segment are grouped to form a *gaggle*, as shown in Fig. 2, which is the least unit for entropy coding in BPE. Due to the different characteristics of DC and AC components, they will be encoded with different techniques.

BPE is designed to process 8×8 blocks. Every 16 consecutive blocks constitute a gaggle and all the blocks in a gaggle are entropy encoded together. Once all the blocks in a segment are processed; the streams obtained for each block in the segment are interleaved in order to produce an encoded segment. Within a segment, BPE processes each coefficient with a bit plane structure. The simplified steps of BPE are shown as follows [19], and for the thorough overview, one is suggested to refer to the CCSDS image data compression standard [9], [10], [11].

- 1). DC coefficients are quantized, and a *Differential Pulse Code Modulation* (DPCM) considering all blocks in the segment is applied, followed by *Golomb-Rice Coding* considering all blocks in a gaggle.

- 2). The remaining DC bit planes not quantized that are greater than the bit-depth of AC if any, are refined.
- 3). The bit depth of AC coefficients from each block is encoded using the same technique as for the quantized DC coefficients.
- 4). From the AC bit depth to the lowest bit plane, DC remaining bit planes are refined, and AC significance and refinement are encoded.
- 5). Stream from each block in the segment are interleaved to construct a quality progressive file.

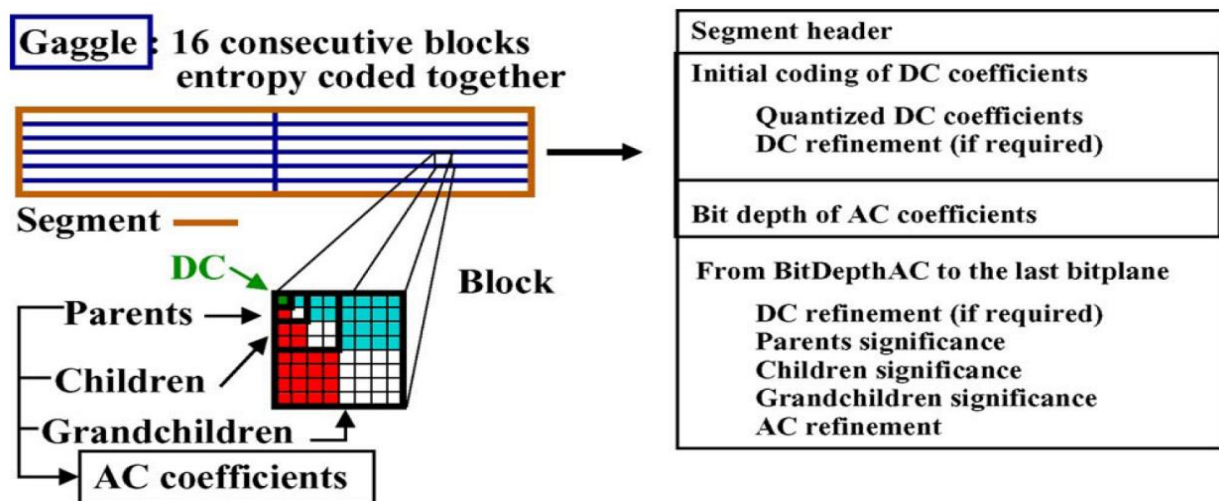


Figure 2: Interleaving procedure and the structure of a block, a gaggle, and a segment

3. Proposed algorithm

In this section, we first apply the MAXSHIFT scheme adopted in JPEG2000 to the CCSDS standard to implement the ROI scalability. However, the coding structures of JPEG2000 and the CCSDS standard are different, so the modification of the coding process is needed. Besides, one of the advantages of the CCSDS standard is the strip-based compression, which can effectively decrease the

hardware cost. It's a pity that MAXSHIFT could just work in the frame-based compression. Therefore, a strip-based compression algorithm with the ROI scalability is also presented in this section.

3.1 MAXSHIFT in the CCSDS standard

In the process of the CCSDS standard, the subband of DC components will be encoded with DPCM. Because the coefficients of ROI are shifted

upwards by $\lceil \log_2 k \rceil$ bits in MAXSHIFT, it causes great differences at the boundary between the DC components of non-ROI and ROI parts and thus it needs more bits to perform DPCM. Different from the MAXSHIFT utilized in [20], in the proposed method, all DC components will be processed as the ROI parts to maintain the efficiency of DPCM. In the encoder, the following steps are performed.

Step 1: Read the input image and the ROI mask and then a 3-level DWT is performed. ALL the coefficients of LL_3 subband are classified as ROI coefficients.

Step 2: Find the maximum value k of the wavelet coefficients of non-ROI parts, and calculate its bit depth $\lceil \log_2 k \rceil$.

Step 3: Shift the wavelet coefficients of ROI parts upwards by $\lceil \log_2 k \rceil$ bits.

Step 4: Encode the value of $\lceil \log_2 k \rceil$ in the header.

Step 5: Perform BPE, and output the final bit stream.

Note that MAXSHIFT for the ROI could just be realized in the case that the entire image is taken as a segment to perform the CCSDS compression since MAXSHIFT implements the ROI scalability by shifting upwards the bit-planes of the relative wavelet coefficients rather than by marking ROI and then compressing marked regions with better quality. Note that in the CCSDS compression standard, each segment is processed at the same compression rate. For example, if an image is partitioned into several strips and certain of them do not contain any ROI information, the shifting scheme will not be performed while processing these strips. Under the condition of compressing each strip of an image at the same compression rate, the quality of these strips will be better than that of non-ROI of other strips containing ROI information and even might be as good as that of ROI of other strips. Thus in the final reconstructed image, the quality of the ROI could not be ensured to be the best. This contradicts the purpose of the ROI scalability.

3.2 Multiple-ROI coding in the CCSDS standard

Here a compression algorithm for realizing the multiple-ROI coding in the strip-based CCSDS standard is presented. In the regrouping process in the CCSDS standard, 1 DC component and 63 AC components are selected from their respective subbands to form a 8×8 block. Here we utilize this characteristic and rearrange these blocks in the order similar to the relative positions of DC components in the LL_3 subband. As shown in Fig. 3, (a) is the original image and (b) is the DWT coefficient matrix after regrouping and rearrangement. It is obvious that the geometric structure of both are similar where the texture in A is more complicated while that in B is smoother. By this observation, for the range of each ROI where the left-top point (x_{lt}, y_{lt}) and the right-bottom point (x_{rb}, y_{rb}) are assigned by users, we re-calculate the corresponding positions (x'_{lt}, y'_{lt}) and (x'_{rb}, y'_{rb}) in the regrouped coefficients by the following equations. As aforementioned, CCSDS compression is based on segments, and therefore each ROI part may be composed of more than one segment.

$$(x'_{lt}, y'_{lt}) = \left(\left\lceil \frac{x_{lt}}{8} \right\rceil \times 8, \left\lceil \frac{y_{lt}}{8} \right\rceil \times 8 \right) \quad (1a)$$

$$(x'_{rb}, y'_{rb}) = \left(\left\lceil \frac{x_{rb}}{8} \right\rceil \times 8, \left\lceil \frac{y_{rb}}{8} \right\rceil \times 8 \right) \quad (1b)$$

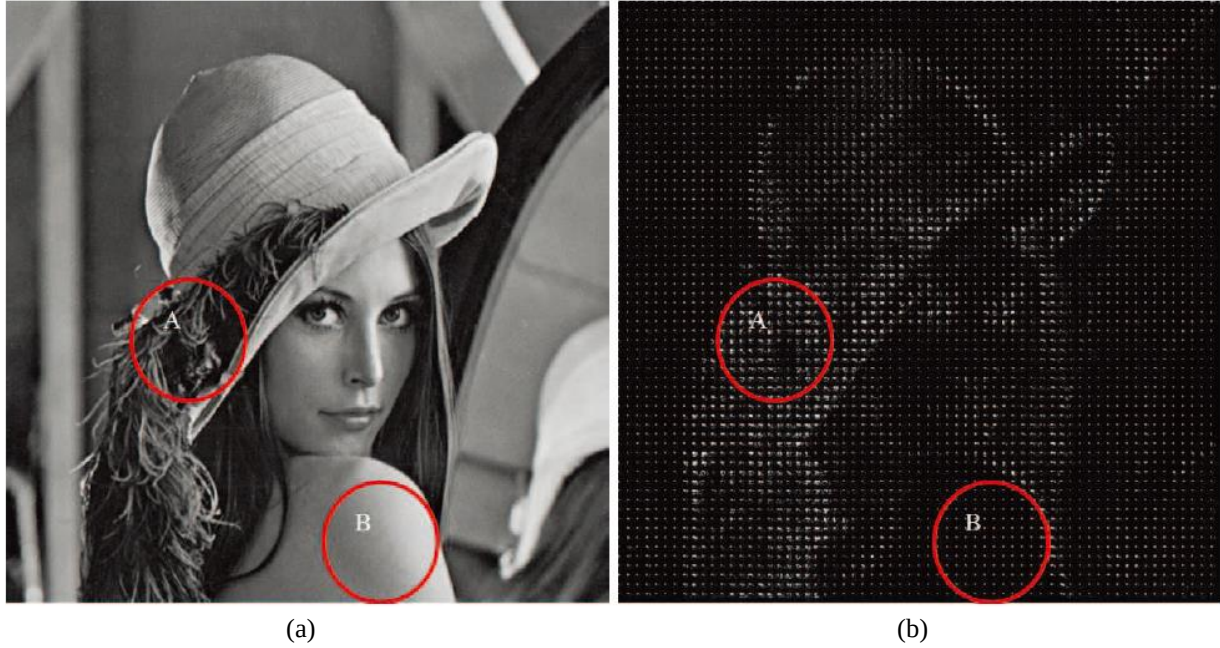


Figure 3: Similar textures: (a) Original image. (b) Rearranged wavelet coefficients.

Prior to BPE, the proper bit allocation of ROI and non-ROI segments is a serious problem that affects the performance of the ROI scalability. Here we propose a novel method that can solve this problem efficiently. For an original image, I_O , we first downsample it to obtain a smaller image, I_S , according to the buffer size. Then the entropy of the ROI segments in I_S is calculated by the following equation, where n is the total number of coefficients in the ROI segments, x_i is the i th coefficient in the ROI segments and $P(x_i)$ is the probability mass function of x_i .

$$H(ROI_S) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i) \quad (2)$$

The entropy $H(ROI_S)$ means the average bits to losslessly encode each coefficient in the ROI segments in I_S , and then we can estimate the entropy, $H(ROI_O)$, of the ROI segments in I_O and further calculate the total necessary bits, B_{ROI} , to losslessly encode the initial ROI segments. Next, to make sure that non-ROI segments could be decoded successfully under any condition of very low bit rate, it is necessary to preserve the bits, $B_{non-ROI}$, for encoding the information of headers of these non-ROI segments in advance. With the target

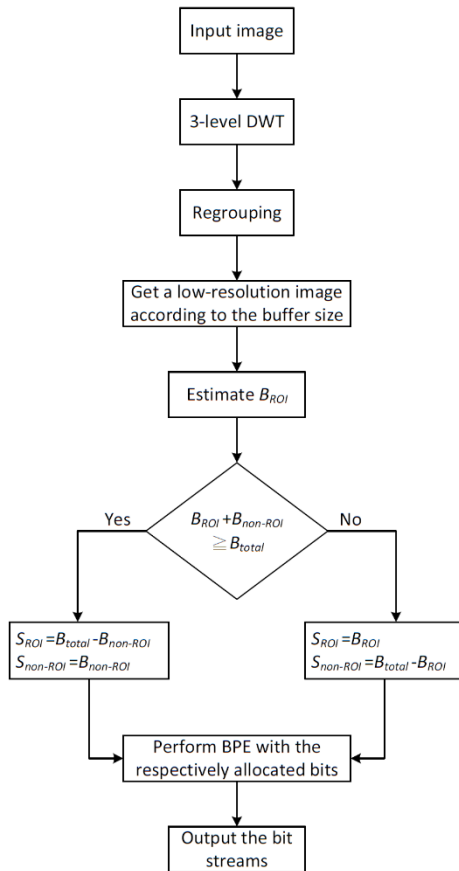
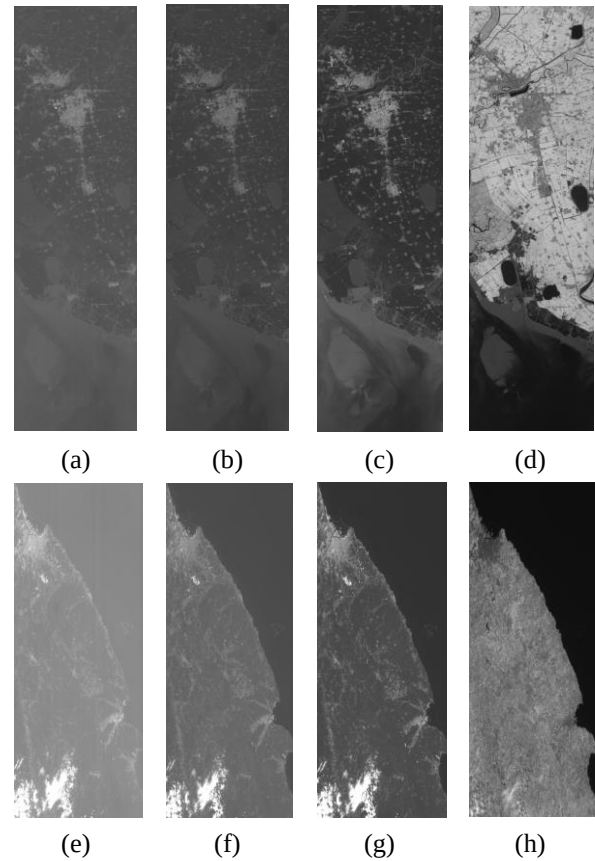
compression rate given by users, we can calculate the total bits, B_{total} , for image compression.

In the process of bit allocation, we first make a comparison between B_{total} and the sum of B_{ROI} and $B_{non-ROI}$. If the latter is greater than or equal to the former, it means that the lossless compression for the ROI segments is impossible. In this case, we try to preserve the image quality of ROI segments as good as possible and all image information in the non-ROI segments will be ignored. Thus we need to preserve the bits, $B_{non-ROI}$, that are necessary while decoding the non-ROI segments, and then the rest bits, $B_{total} - B_{non-ROI}$, are all distributed to the ROI segments. On the contrary, if the former is greater than the latter, that is, the ROI segments can be losslessly compressed, we allocate bits to the ROI segments prior to the non-ROI ones. The details of bit allocation are shown in Algorithm 1.

Algorithm 1 Bit allocation algorithm**Input:** An ROI mask and the given compression rate C **Output:** S_{ROI} bits for the ROI segments and $S_{non-ROI}$ bits for the non-ROI segments

- 1: An original image is downsampled according to the available cache and calculate $H(ROI_s)$ by the modified ROI mask;
- 2: Estimate $H(ROI_o)$ and calculate B_{ROI} ;
- 3: Calculate $B_{non-ROI}$;
- 4: Calculate B_{total} by the given compression rate C ;
- 5: **if** $B_{ROI} + B_{non-ROI} \geq B_{total}$ **then**
- 6: $S_{ROI} = B_{total} - B_{non-ROI}$;
- 7: $S_{non-ROI} = B_{non-ROI}$;
- 8: **else**
- 9: $S_{ROI} = B_{ROI}$;
- 10: $S_{non-ROI} = B_{total} - B_{ROI}$;
- 11: **end if**
- 12: **return** S_{ROI} and $S_{non-ROI}$;

In the end, BPE is performed to encode the ROI and non-ROI segments to fit the respective allocated bits. Fig. 4 is the complete flowchart of the proposed algorithm.

**Figure 4:** Flow chart of the proposed algorithm**4. Simulation results****Figure 5:** Test images captured with different frequencies

There are two test image sets obtained from Nation Space Organization (NSPO) in this experiment. There are four 8-bit satellite images captured with rays at different spectra in each image set. The dimension of these satellite images in set I, as shown in Fig. 5 (a)-(d), is 3000×10440 , and that of the images in set II, as shown in Fig. 5 (e)-(h), is 3000×7800 . Note that the last images in both sets are more complicated than others. For example, as we can see in Fig. 5, there are more textures in (d) than that in (a), (b) and (c). It means that the information contained in (d) is more useful and more important.

ROIs assigned by users are usually the most important or the most concerned regions in an image. Thus the quality of these ROIs is always supposed to be better than that of other regions in an image whatever the given compression rate is. Fig. 6 shows the performance of the ROI scalability at different compression rates (or bits per pixel, bpp) where ROIs, in the white rectangles, are randomly assigned. It is obvious that the visual quality of these ROIs is always better than that of other regions in the image. Besides, we can clearly find the differences in the black rectangle at different compression rates. When the compression rate is great (bpp is small), as shown in Fig. 6(b), where the compression rate is 40, in the black rectangle the ROI is a little blurred while the non-ROI is very terrible. With the decreasing of the compression rate, the quality of these ROIs becomes lossless and furthermore, as shown in Fig. 6(f), where the compression rate is 8, the visual quality of the non-ROI also becomes much better.

Note that since the ROI is usually the most concerned region, in our experiment, only the PSNR of the ROI is concerned. For each test image set at a certain compression rate, we performed the ROI compression 10 times, and at each time the ROIs were randomly selected. Table I and II show the simulation results of PSNR in the full-frame mode of

the proposed algorithm, modified MAXSHIFT, JPEG2000 and the scaling-based utilized in [20] where the parameter of bits shifted upwards is set as 4. The following equation is adopted to calculate the PSNR of n ROIs in a test image, where Q is the maximum possible value of the samples (here Q is set as 255 since the bit-depth of the test images is 8), $u(i, j)$ is the distortion between the original pixel value and the reconstructed pixel value, and W_k and H_k represent the width and the height of the k th ROI in a test image.

$$PSNR = \frac{1}{n} \sum_{k=1}^n 10 \log_{10} \frac{Q^2}{\left(\frac{1}{W_k H_k} \sum_{i=0}^{W_k-1} \sum_{j=0}^{H_k-1} u^2(i, j)\right)} \quad (3)$$

It is obvious that the PSNR of the modified MAXSHIFT in the CCSDS standard is the worst of these results. Taking the whole LL subband as a part of ROI increases the cost of entropy coding and thus greatly decreases the compression performance. The performance of the scaling-based ROI algorithm proposed in [20] is also decreased by the cost of the ROI mask. In the cases of small bpp, the performance of the proposed method is not so good as that of JPEG2000 because of the segment-based structure of the original CCSDS standard. The burdens of these segment headers containing necessary information for decoding become relatively large while the bpp is very small. However, in the general range of the compression rate, from 7.5 to 15, the proposed method performs better than JPEG2000. Besides, as presented in Table II, the PSNR of the proposed method is better in the complicated image (h) in set II although the average PSNR is not so good as that of JPEG2000.

One of the contributions of the proposed method is the strip-based ROI coding, and in Table III, the PSNR performance of the different numbers of scan lines of strip-based ROI coding is presented. We can find that the average PSNR of the strip-based mode is about 3-5dB lower than that of the frame-based mode, as shown in Fig. 7. This phenomenon is caused by the increasing of the segments. When the strip-based ROI coding is performed to meet the limitation of the

buffer size, an image has to be partitioned into several non-overlapped strips. In this case, more headers are needed and thus the compression performance is affected. However, once the

strip-based ROI coding could be performed as a compression candidate, the design of the buffer size becomes more flexible and thus the cost of the hardware could be greatly decreased.

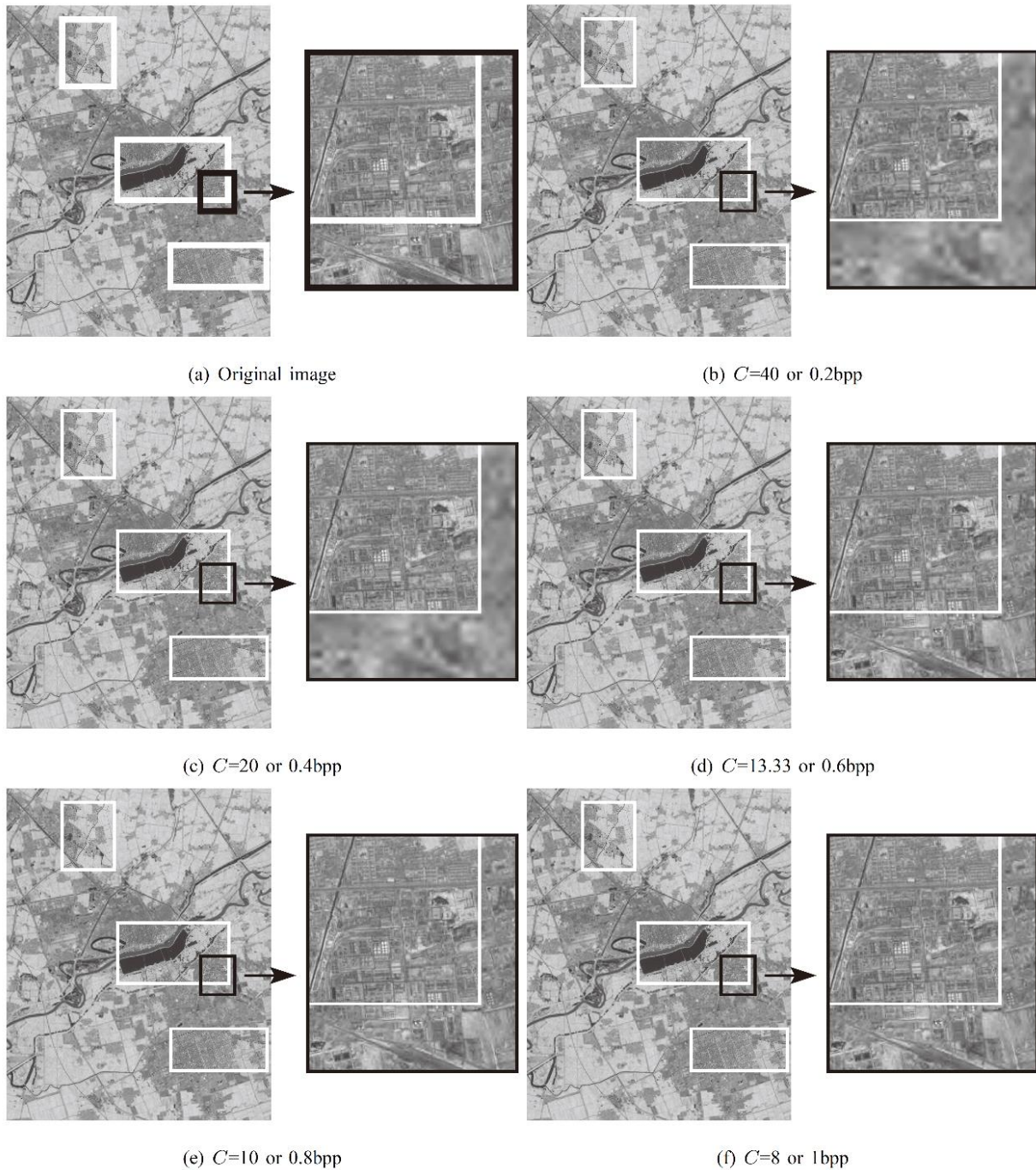


Figure 6: Performance of the ROI scalability at different compression rates

Table 1: PSNR comparison of test image set I

		BPP					
		0.2	0.4	0.6	0.8	1.0	1.2
Proposed method	B1	37.5992	41.7863	44.6476	46.9019	48.8253	50.5213
	B2	36.0349	40.1008	43.0368	45.3390	47.4255	49.2691
	B3	32.1372	36.4374	39.5421	42.1524	44.3806	46.4603
	B4	27.079	30.5976	33.1923	35.3440	37.3948	39.3920
	Average	33.2126	37.2305	40.1047	42.4343	44.5066	46.4107
Modified MAXSHIFT	B1	36.6493	40.0188	42.3124	44.5399	46.4030	47.9745
	B2	35.1142	38.4608	40.7381	42.6678	44.8428	46.5341
	B3	31.0395	34.8469	37.3156	39.3887	41.2743	43.2091
	B4	26.6974	29.5556	31.4724	33.2341	34.5547	35.9467
	Average	32.3731	35.7025	37.9597	39.9576	41.7687	43.4161
Scaling of 4 bits [20]	B1	36.9254	40.3278	42.6415	44.8478	46.7003	48.2587
	B2	35.4042	38.7580	40.9875	42.9745	45.1058	46.8472
	B3	31.3452	35.1459	37.6874	39.6975	41.5428	43.5109
	B4	26.9798	29.8655	31.7896	33.5412	34.8749	36.2571
	Average	32.6637	36.0243	38.2765	40.2653	42.0560	43.7185
JPEG 2000	B1	38.5487	41.4209	43.7141	45.8842	48.0747	50.4192
	B2	37.0836	39.8110	42.1209	44.2187	46.2071	48.5216
	B3	34.0271	37.1171	39.4354	41.5081	43.4515	45.5603
	B4	28.3132	31.0375	33.1583	34.6891	36.3391	37.9415
	Average	34.4932	37.3466	39.6072	41.5750	43.5181	45.6107

5. Conclusions

Image compression with the ROI scalability is becoming popular. For satellite image compression that deals with the huge amount of data captured by sensors, the image compression standard proposed by CCSDS provides an excellent tradeoff between performance and complexity for medium and high

target bit rates. However, the compression rate is the only one factor concerned in the original CCSDS standard. JPEG2000 is a popular and powerful image compression method. It also provides an excellent ROI scalability. It's a pity that ROI coding in JPEG2000 just performs well in the frame-based

mode rather than in the strip-based mode; thus, the cost of buffer and hardware is a heavy burden.

To overcome these limitations, a strip-based image compression algorithm based on the CCSDS standard providing the ROI scalability is proposed in this study. From the simulation results, it is obvious that the performance of the proposed algorithm is competitive to JPEG2000, and moreover, the performance of the proposed algorithm is still good

in the strip-based mode. The proposed algorithm not only keeps the advantages of the CCSDS standard, but also provides the strip-based ROI scalability that could not be realized in JPEG2000. The compatibility with existing CCSDS hardware systems and the supply of the strip-based ROI scalability make it easier to be implemented in hardware design, and furthermore, the cost of the image processing hardware system could be effectively decreased.

Table 2: PSNR comparison of test image set II

		BPP					
		0.2	0.4	0.6	0.8	1.0	1.2
Proposed method	B1	38.7264	42.8956	45.7184	48.1300	50.3788	52.5065
	B2	37.3523	41.1588	43.9326	46.1618	48.6361	50.6543
	B3	33.8704	38.2239	41.5043	44.2287	46.8517	49.3164
	B4	29.7526	33.0458	35.2986	37.8934	39.8726	42.6527
	Average	34.9254	38.8310	41.6135	44.1035	46.4348	48.7825
Modified MAXSHIFT	B1	36.7023	39.9659	42.1769	44.2930	45.8526	47.4584
	B2	35.4764	38.5169	40.3704	42.2941	44.0800	45.4844
	B3	31.8426	35.4810	37.7861	39.6977	41.6441	43.5957
	B4	27.9172	30.6453	32.4460	33.7185	34.9291	36.2594
	Average	32.9846	36.1523	38.1948	40.0008	41.6264	43.1995
Scaling of 4 bits [20]	B1	36.9923	40.3569	42.4810	44.5903	46.1652	47.7485
	B2	35.7654	38.8096	40.6840	42.5921	44.3811	45.7952
	B3	32.1654	35.7789	38.0816	39.9879	41.9374	43.9013
	B4	28.2271	30.9534	32.7604	34.0158	35.2941	36.5614
	Average	33.2986	36.4747	38.5018	40.2965	41.9445	43.5016
JPEG 2000	B1	40.2695	44.0961	46.6346	48.2572	50.0756	52.6695
	B2	38.3796	41.7717	44.0196	45.9659	47.3696	49.7055
	B3	35.7875	39.8787	42.9503	45.3509	47.0525	49.1882
	B4	30.3396	33.0362	35.2210	37.3973	39.3251	41.7436
	Average	36.1932	39.6956	42.2064	44.2428	45.9557	48.3267

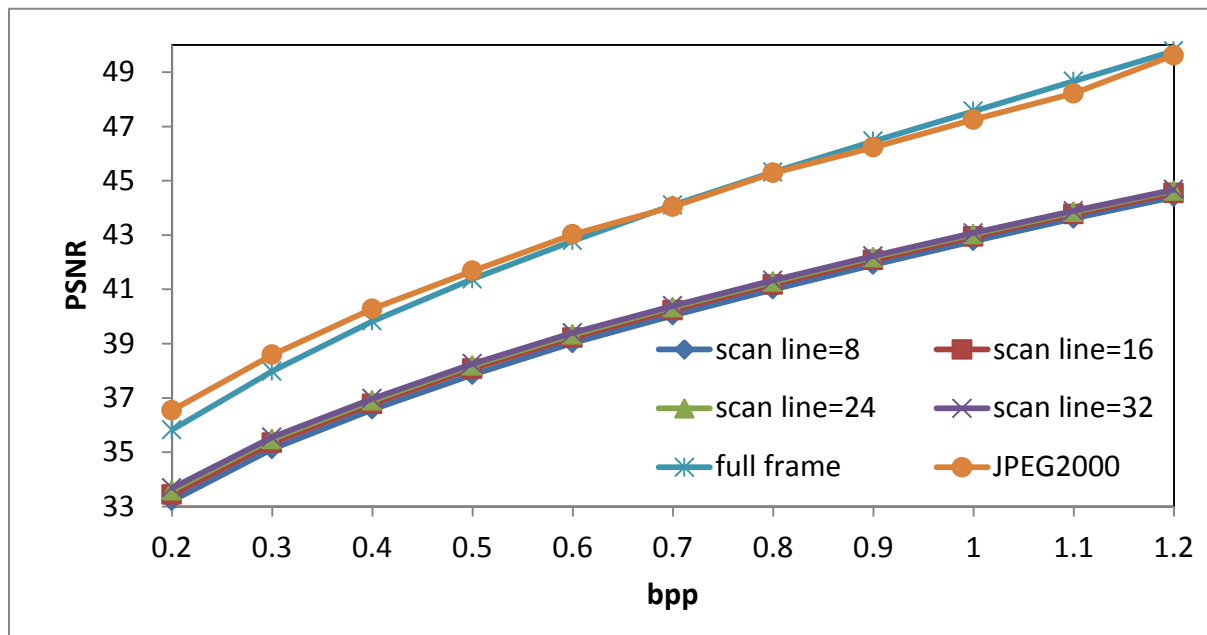


Figure 7: Average PSNR comparison of strip-based ROI containing different scan lines and JPEG2000 in set I.

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